

TURINSK, Russian Federation

WMO: 282550

Lat: 58.050N

Lon: 63.683E

Elev: 103

StdP: 100.10

Time Zone: 5.00 (E05)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.6	(c) -30.1	(d) -35.6	(e) 0.1	(f) -32.5	(g) -33.0	(h) 0.2	(i) -30.0	(j) 9.0	(k) -8.8	(l) 7.9	(m) -8.3	(n) 1.4	(o) 260	(p) 0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.5	(c) 29.6	(d) 21.0	(e) 27.8	(f) 20.1	(g) 25.9	(h) 19.0	(i) 22.2	(j) 27.8	(k) 21.0	(l) 26.4	(m) 19.8	(n) 24.7	(o) 1.9	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 20.1	(b) 15.0	(c) 25.8	(d) 19.0	(e) 14.0	(f) 24.3	(g) 17.9	(h) 13.0	(i) 22.5	(j) 65.5	(k) 27.8	(l) 61.4	(m) 26.4	(n) 57.3	(o) 24.8	(p) 25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.6	(b) 6.4	(c) 5.6	(e) -34.9	(f) 32.5	(g) 3.2	(h) 1.3	(i) -37.2	(j) 33.5	(k) -39.1	(l) 34.2	(m) -40.9	(n) 34.9	(o) -43.3	(p) 35.9		
(a) 7.6	(b) 6.4	(c) 5.6	(e) -35.0	(f) 23.6	(g) 3.2	(h) 1.1	(i) -37.3	(j) 24.3	(k) -39.2	(l) 25.0	(m) -41.0	(n) 25.6	(o) -43.3	(p) 26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-15.9	-13.6	-4.9	3.8	11.2	16.2	18.2	15.6	9.8	2.9	-7.1	-13.7		
	DBStd	13.44	7.69	7.62	5.16	5.57	5.29	4.51	4.05	4.13	4.60	5.12	7.71	8.17		
	HDD10.0	3721	804	662	462	197	51	6	1	3	60	227	514	736		
	HDD18.3	6090	1062	895	720	436	227	92	54	107	259	479	764	994		
	CDD10.0	779	0	0	0	11	88	192	254	176	53	6	0	0		
	CDD18.3	107	0	0	0	0	6	28	49	21	2	0	0	0		
	CDH23.3	1069	0	0	0	2	93	293	477	184	19	0	0	0		
	CDH26.7	279	0	0	0	0	14	81	130	52	2	0	0	0		
(14) Wind	WSAvg	2.2	2.3	2.2	2.4	2.5	2.4	2.0	1.5	1.7	2.0	2.4	2.4	2.3		
(15) Precipitation	PrecAvg	547	30	23	26	30	49	66	71	78	59	42	38	34		
	PrecMax	763	58	56	63	72	119	132	155	190	106	94	71	66		
	PrecMin	407	4	3	1	1	11	30	21	17	10	4	8	11		
	PrecStd	84	15	12	15	21	28	26	34	40	29	22	17	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	3.2	8.6	22.5	28.4	31.4	32.2	31.3	25.7	18.4	7.1	2.4		
		MCWB	0.5	0.8	3.9	13.5	17.0	21.4	22.2	21.9	17.7	12.3	5.1	1.0		
	2%	DB	-1.0	1.2	5.3	17.9	25.9	29.0	29.7	27.8	22.3	15.0	4.8	0.8		
		MCWB	-2.1	-0.3	2.0	11.0	16.1	20.2	21.5	20.5	16.3	10.7	3.5	-0.4		
	5%	DB	-2.9	-0.8	3.5	14.8	23.6	26.8	28.0	25.0	19.2	12.2	3.1	-1.1		
		MCWB	-3.7	-2.2	0.8	9.0	15.1	19.5	20.9	19.3	14.8	9.1	1.9	-2.1		
	10%	DB	-5.0	-2.8	2.0	12.2	20.9	24.5	26.2	22.8	16.6	10.0	1.6	-3.0		
		MCWB	-5.8	-4.1	-0.2	7.4	13.9	18.4	20.1	18.2	13.2	7.8	0.5	-3.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	1.4	4.3	13.7	18.7	23.0	23.7	22.9	18.8	13.0	5.6	1.2		
		MCDB	1.6	2.8	7.7	20.6	25.0	29.1	29.6	29.7	23.5	16.7	6.9	2.2		
	2%	WB	-1.8	-0.2	2.4	11.4	17.0	21.3	22.5	21.1	16.7	10.9	3.8	-0.3		
		MCDB	-1.1	1.2	4.9	17.1	24.6	27.4	28.1	26.3	21.4	14.2	4.9	0.7		
	5%	WB	-3.7	-2.1	1.2	9.4	15.5	20.1	21.5	19.8	15.1	9.3	2.1	-1.9		
		MCDB	-2.9	-0.9	3.1	14.2	22.4	25.9	26.8	24.2	18.8	12.1	3.0	-1.2		
	10%	WB	-5.8	-4.1	0.1	7.6	14.1	18.8	20.4	18.5	13.5	7.8	0.6	-3.8		
		MCDB	-5.2	-2.9	1.8	11.8	20.1	23.4	25.3	22.1	16.1	9.9	1.4	-3.0		
(35) Mean Daily Temperature Range	MDBR		7.0	8.3	9.1	9.8	11.4	10.8	10.5	9.6	8.6	6.8	6.0	6.4		
	5% DB	MCDBR	6.9	7.4	9.3	13.4	15.4	14.0	13.3	12.9	12.1	10.5	6.1	6.0		
		MCWBR	6.7	6.2	6.7	8.2	8.0	7.0	6.2	6.5	7.0	6.7	5.2	5.6		
	5% WB	MCDBR	7.0	6.9	8.1	12.6	14.3	12.8	12.4	11.9	11.5	10.0	5.9	5.8		
		MCWBR	6.9	5.9	6.2	8.1	8.0	7.1	6.4	6.4	7.0	6.6	5.2	5.6		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.311	0.366	0.406	0.401	0.402	0.418	0.429	0.380	0.344	0.303	0.274		
	taud		2.036	2.034	1.953	2.093	2.260	2.318	2.273	2.214	2.314	2.277	2.134	1.995		
	Ebn at Noon		526	688	747	793	831	836	810	766	750	661	498	397		
	Edh at Noon		56	90	131	135	122	117	120	119	93	72	51	43		
(44) All-Sky Solar Radiation	RadAvg		0.61	1.56	3.00	4.20	5.09	5.54	5.38	4.07	2.48	1.28	0.58	0.35		
	RadStd		0.05	0.12	0.18	0.40	0.45	0.39	0.46	0.40	0.25	0.20	0.08	0.05		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page